

5. route.py Development

ComicCraft Project Documentation

Purpose: Provides a focused explanation of the routes.py milestone, its endpoints, input handling, AI integration, and error-handling approach.

5.1 Purpose of routes.py

The source document describes routes.py as the main application logic layer for FastAPI routing. It receives user input, invokes the AI generation functions, builds the comic layout, and prepares the result for the frontend or API client.

5.2 Main Routes

Route	Purpose
/	Loads the homepage index.html.
/generate	Processes form input and generates the comic preview.
/generate-comic/json	Accepts JSON input and returns layout data and PDF path.
/export-success	Displays export/download success confirmation.
/test-image	Tests image generation independently using a custom prompt.

5.3 Form Inputs

- story prompt
- main character name
- setting
- story tone
- art style

The /generate route receives these values through FastAPI Form(...) parameters.

5.4 JSON API

The /generate-comic/json route receives a JSON body validated with the documented PromptRequest Pydantic model. It follows a workflow similar to the form-based route and returns comic layout information and the generated PDF path.

5.5 AI Workflow Inside the Routes

```
/generate or /generate-comic/json
↓
generate_outline()
↓
generate_story()
↓
for each panel: generate_image()
↓
build_comic_layout()
↓
save_pdf()
↓
TemplateResponse or JSON response
```

5.6 Error Handling

The documented route handlers use FastAPI `HTTPException` and `try-except` blocks to handle failures such as API errors and JSON decoding problems. Generated content is passed dynamically to the Jinja2 templates or returned to API clients.